

Adaptive Game Model - Technical Report

Event-based tactical decision intelligence: methodology, validation, and reproducibility

1. Problem & modelling decision

We estimate which build-up adjustment should be used against a specific opponent structure - not which team wins. Every possession is labelled with a 0-5 outcome scale and converted to expected possession value (EPV) in goal-probability units. The value model (XGBoost + isotonic calibration) is trained on ex-ante features; tactical states are discovered by clustering and named by football prototypes; adjustment values are estimated causally with stabilised IPW and a doubly-robust (AIPW) augmentation, with match-clustered bootstrap confidence intervals.

2. Data & preprocessing

Source: synthetic | 24 matches, 1955 possessions.

Check	Severity	Count	Message
duplicate_events	info	0	No duplicate events
impossible_coordinates	info	0	All coordinates within pitch
missing_player_ids	info	0	0/6906 on-ball events lack player_id (0.0%)
invalid_timestamps	info	0	All timestamps valid
team_name_consistency	info	0	Team names consistent with manifest
competition_mismatch	info	0	All events map to manifest matches
outliers	info	0	No outlier values flagged
leakage	info	0	No match-level leakage between train and test

3. Possession value model

Model	RMSE	MAE	R2
xgboost_calibrated	0.0301	0.0222	0.118
xgboost_raw	0.0300	0.0216	0.120
linear	0.0277	0.0202	0.253
elo_only	0.0323	0.0255	-0.016
league_average	0.0322	0.0257	-0.010

4. Calibration & temporal validation

Out-of-time calibration ECE = 0.0026 (MSE 0.00090). The model is evaluated strictly chronologically: train on earlier matches, validate and test on later matches. No action from a match appears in more than one split (leakage check passes).

5. Tactical states

Opponent high press - 1750 sequences, baseline EPV 0.015.

Final-third attack vs compact defence - 205 sequences, baseline EPV 0.058.

6. Adjustment value estimates

State	Adjustment	Uplift	CI	Confidence
Final-third attack vs compact defence	s_third_man_combination	+0.013	[+0.008, +0.019]	high
Final-third attack vs compact defence	s_reduce_risky_central_passes	+0.010	[-0.002, +0.020]	medium
Final-third attack vs compact defence	s_increase_winger_width	+0.004	[-0.012, +0.013]	medium
Final-third attack vs compact defence	s_aggressive_counterpress	+0.001	[-0.010, +0.014]	medium
Opponent high press	s_third_man_combination	+0.011	[+0.009, +0.013]	high
Opponent high press	s_increase_winger_width	+0.006	[+0.000, +0.012]	high
Opponent high press	s_reduce_risky_central_passes	-0.002	[-0.004, +0.001]	medium
Opponent high press	s_mid_block_retreat	-0.006	[-0.007, -0.005]	high
Opponent high press	s_aggressive_counterpress	-0.010	[-0.013, -0.007]	high

7. Error analysis

Three worst failures:

FC Azul Marino vs Villarreal Amarillo (min 37): actual 0.138 vs predicted 0.011 - Model under-predicted value in unknown: the possession created a dangerous action (shot assist / box entry) that the ex-ante features could not foresee.

Getafe Azulon vs FC Azul Marino (min 15): actual 0.111 vs predicted 0.009 - Model under-predicted value in unknown: the possession created a dangerous action (shot assist / box entry) that the ex-ante features could not foresee.

FC Azul Marino vs Sociedad Txuri (min 31): actual 0.112 vs predicted 0.011 - Model under-predicted value in unknown: the possession created a dangerous action (shot assist / box entry) that the ex-ante features could not foresee.

8. Ablation & sensitivity

Feature group	RMSE	MAE	R2
full	0.0291	0.0208	0.175
no_elo	0.0290	0.0205	0.179
no_match_context	0.0290	0.0208	0.180
no_location	0.0311	0.0225	0.058

9. Known limitations

- Event-based: no player tracking data is used.
- Adjustment labels are transparent rule-based proxies of coaching concepts.
- Estimates are historical expectations with uncertainty, not guarantees.
- Applicability limited to opponents/leagues represented in training data.
- Small samples yield low-confidence recommendations by design.

10. Reproducibility

Run `python -m src.pipeline.run_all` to reproduce all artefacts. Data manifest with SHA-256 fingerprints lives in `data/processed/data_manifest.json`. Random seed fixed (42); synthetic fallback is deterministic for offline CI.